A Mathematical Solution to the AI Alignment Problem: Topological Constraints on Action Distributions with Progressive Verification

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Abstract

We give an *ab initio* (first-principles) mathematical construction for AI alignment that is explicitly **decoupled from training data quality** (A2): the base model may be weak, biased, or even strategically misaligned, yet the deployed system is aligned *by construction* via an external constraint wrapper and monitor. The core move is to treat a deployed AI system as inducing a probability law over infinite action–observation trajectories, and to enforce alignment as a *topological membership condition* $\mu \in \mathcal{S}$ in a well-behaved class of probability measures. We introduce **progressive outputs** (A3): filtration-aligned partial outputs that make nondeterminism non-hidden, so violations manifest as measurable *distributional drift* in trajectory space, detectable by sequential tests.

Formally, we (i) define an infinite-dimensional path space Ω and $\mathcal{P}(\Omega)$ endowed with weak/Wasserstein topologies; (ii) encode safety, legality, and corrigibility as scalar *risk functionals* $\Phi_i : \mathcal{P}(\Omega) \to \mathbb{R}$ (A4); (iii) prove existence of *aligned projections* of raw behavior laws onto $\mathcal{S}$ under explicit Wasserstein regularity and coercivity conditions (A5); (iv) bridge infinite-dimensional theory to finite logs via conformal/PAC certification bounds (A6); and (v) extend to multi-agent settings using equilibrium existence on locally convex spaces.

Scope restriction: we focus on *information-work* systems (analysis, reasoning, decision support, office workflows) without direct physical actuation. Excluding physical systems improves feasibility and auditability but can invite criticism; we therefore position the framework as foundational and extensible to embodied systems via a shielded physical interface layer (Related Work, Conclusion).

1 Scope and Design Axioms (A1–A6)

This paper is **not** about making models more capable. It is about making deployed systems *aligned*. We adopt six first-principles axioms and reference them explicitly throughout.

A1: Alignment is a property of deployed behavior, not internal narratives.

A2: Decoupling from data quality: poor data can make a model incompetent, but should not make it unsafe.

A3: Nondeterminism must not be a hiding place: emit *progressive outputs* so hidden intent/hallucinations/reward hacking are detectable as distributional anomalies.

A4: Constraint minimalism: constrain *actions and action distributions* rather than demand a complete theory of internal "values".

A5: Infinite-dimensional readiness: reason in probability spaces over trajectory spaces to accommodate unknown future capabilities without changing foundations.

A6: Observable alignment verification: all safety arguments rest upon externally verifiable facts (logs, certificates, distributions), not trust in internal mechanisms.

Scope boundary strength/weakness (A2, A6). We restrict to systems whose actions are *informational* and whose external effects are mediated by auditable digital interfaces. This strengthens focus and feasibility. To mitigate "what about embodied AI?" objections, we treat this work as the *foundational layer* that can later be composed with a formally verified physical shield.

1.1 Positioning Against Related Work (A1, A4, A6)

This framework complements (does not replace) leading alignment lines:

- **Value-shaping** training (RLHF, Constitutional AI) aims to shape internal preferences [9, 10]. Here the trained model may be adversarial; alignment is enforced externally (A1, A6).
- **Debate/oversight** seeks truthful reasoning via interaction [11]. We instead rely on progressive, certificate-bearing outputs plus non-AI verifiers (A3, A6).
- **CMDPs and shielding** [27, 28] appear as special cases of constrained behavior laws and safety wrappers (A4, A6).

2 Formalizing Alignment as Topology on Laws of Behavior (A1, A5)

2.1 Trajectory space and induced laws (A1)

Let $\mathcal{O}$ be an observation space and $\mathcal{A}$ an action space. Define

$$\Omega := (\mathcal{O} \times \mathcal{A})^{\mathbb{N}}$$

with product σ-algebra $\mathcal{F}$. A deployed system induces a probability measure $\mu \in \mathcal{P}(\Omega)$. Alignment is a *membership constraint* $\mu \in \mathcal{S} \subseteq \mathcal{P}(\Omega)$.

Assumption 1 (Well-behaved base spaces (A5))**.** Assume $\mathcal{O} \times \mathcal{A}$ is Polish. Then Ω is Polish under standard product metrics, and $\mathcal{P}(\Omega)$ is well-behaved under weak convergence [19]. For quantitative bounds we work in $\mathcal{P}_2(\Omega)$ with Wasserstein W_2 [20, 21].

2.2 Progressive outputs as filtrations (A3, A6)

Let $(\mathcal{F}_t)_{t\in\mathbb{N}}$ be the natural filtration generated by prefixes $\omega_{0:t}$. **Progressive outputs** mean at each t the system emits a structured partial output y_t that is $\mathcal{F}_t$-measurable and sufficient to run monitors.

Definition 1 (Progressive output contract (A3))**.** At each step t, the system emits

$$y_t = (p_t, c_t, v_t, \ell_t),$$

where p_t is a partial plan/summary, c_t is a verifier payload, $v_t \in \{\texttt{pass}, \texttt{fail}, \texttt{unknown}\}$ is produced by a *non-AI* checker, and ℓ_t is a cryptographic commitment for tamper evidence [44, 45, 46].

3 Alignment Constraints as Risk Functionals (A4)

3.1 Codex constraints as expectations (A4, A6)

Let $r_i : \Omega \to [0,1]$ be a measurable risk score for Law i. Define

$$\Phi_i(\mu) := \mathbb{E}_{\omega\sim\mu}[r_i(\omega)] = \int_\Omega r_i(\omega)\, d\mu(\omega).$$

Critical laws use $\Phi_i(\mu) = 0$ (or $\le \varepsilon_i$); soft laws use tolerances $\varepsilon_i > 0$.

Definition 2 (Aligned feasible set (A4))**.** Given tolerances $\varepsilon_i \ge 0$, define

$$\mathcal{S} := \{\mu \in \mathcal{P}(\Omega) : \Phi_i(\mu) \le \varepsilon_i \ \forall i, \text{ and budget/disclosure constraints hold}\}.$$

Geometric constraints (A4, A6). An alternative (complementary) route is to constrain the *interaction representation space* itself so that both human inputs and AI outputs must lie near certified "anchor" sets and admissible cones, enforced by metric/proximal projections. This can reduce manual per-law specification of r_i for subjective laws by pushing part of the burden into space design and change-controlled anchor libraries; see Appendix C.

3.2 Information minimization and retention (A2, A6)

Let B_t be retained sensitive bits at time t and $\gamma \in (0,1)$.

$$\Phi_{\text{ret}}(\mu) := \mathbb{E}_\mu\Big[\sum_{t\ge 0}\gamma^t B_t\Big] \le \tau.$$

This is consistent with data minimization principles [15]. An entropy-based variant uses Shannon entropy [42].

3.3 Energy/compute budgets (A2, A6)

Let E_t be energy and C_t compute at step t:

$$\Phi_{\text{energy}}(\mu) := \mathbb{E}_\mu\Big[\sum_{t\ge 0}\gamma^t E_t\Big] \le E_{\text{max}}, \quad \Phi_{\text{compute}}(\mu) := \mathbb{E}_\mu\Big[\sum_{t\ge 0}\gamma^t C_t\Big] \le C_{\text{max}}.$$

Resource control is both safety- and governance-relevant [41].

3.4 Reward tampering and deception proxies (A3, A6)

Let Rewards_t be the empirical reward distribution and $\widehat{\text{Rewards}}_t$ a reference:

$$\Phi_{\text{reward}}(\mu) := \sup_{t\ge T_0} \mathrm{D}_{\text{KL}}(\text{Rewards}_t \| \widehat{\text{Rewards}}_t),$$

using KL divergence [43].

3.5 Distributional drift functional (A3, A6)

Let μ_t be the induced prefix law (estimated from logs) and $\widehat{\mu}_t$ a compliant baseline. For a metric d (e.g., W_2),

$$\Phi_{\text{drift}}(\mu) := \sup_{t\ge 1} d(\mu_t, \widehat{\mu}_t).$$

Sequential change detection (SPRT/CUSUM) is standard [24, 25, 26].

4 Existence of Feasible Alignment and Safe Projection (A2, A5)

4.1 Non-emptiness via safe no-op (A2)

Proposition 1 (Non-emptiness (A2))**.** Assume there exists a "safe no-op" action $a_\varnothing \in \mathcal{A}$ with no external actuation. Then $\mathcal{S} \neq \emptyset$.

Proof. The policy that always outputs $a_\varnothing$ induces $\mu_\varnothing$ with zero external harm by construction, hence $\Phi_i(\mu_\varnothing) = 0 \leq \varepsilon_i$ for critical laws, and budget constraints hold.

4.2 Tightness and coercive tail penalties (A5)

The existence theory needs operational tightness. A concrete engineering-compatible example is a coercive exponential tail penalty:

$$\Phi_{\text{tail}}(\mu) := \mathbb{E}_\mu\Big[\sum_{t\geq 0} \gamma^t \exp(\eta E_t)\Big] \leq K,$$

which induces exponential decay on resource tails and supports $\mathcal{P}_2(\Omega)$ control.

Remark 1 (Finite-horizon evasion probability bounds (A5, A6))**.** If the monitor enforces a moment bound $\mathbb{E}[\exp(\eta E_t)] \leq M$ (as in the tail penalty), then Markov/Chernoff implies

$$\mathbb{P}(E_t \geq u) \leq M e^{-\eta u}.$$

Under stronger variance/mgf assumptions (e.g., sub-Gaussian deviations around the mean), sharper tails of the form $\mathbb{P}(E_t - \mathbb{E}[E_t] \geq u) \leq \exp(-u^2/(2\sigma^2))$ follow from standard exponential concentration arguments [40]. These bounds tighten control of *resource* evasion but do not, by themselves, rule out energy-orthogonal harms or strategic deception (Limitations).

4.3 Rigorous Wasserstein projection theorem (A2, A5)

Let μ^{raw} be the base system behavior law.

Theorem 1 (Wasserstein projection onto $\mathcal{S}$ (A2, A5))**.** Let $(\mathcal{O}\times\mathcal{A}, \rho)$ be compact metric and endow Ω with

$$d_\infty(\omega, \omega') = \sum_{t\geq 0} 2^{-(t+1)} \rho(\omega_t, \omega'_t).$$

Let $\mathcal{P}_2(\Omega)$ be laws with finite second moment under d_∞. Assume:

1. $\mu^{\text{raw}} \in \mathcal{P}_2(\Omega)$,
2. each Φ_i is W_2-continuous on $\mathcal{P}_2(\Omega)$, hence $\mathcal{S} = \{\mu : \Phi_i(\mu) \leq \varepsilon_i \ \forall i\}$ is W_2-closed,
3. Φ_{ret} and Φ_{energy} are W_2-continuous and *coercive*: $\Phi_{\text{ret}}(\mu)+\Phi_{\text{energy}}(\mu) \to +\infty$ whenever $\mathrm{W}_2(\mu, \delta_{\omega_0}) \to \infty$ for a fixed $\omega_0 \in \Omega$,
4. $\mathcal{S} \neq \emptyset$ (Proposition 4.1).

Define the objective

$$J(\mu) = \mathrm{W}_2(\mu, \mu^{\text{raw}})^2 + \lambda\, \Phi_{\text{ret}}(\mu) + \lambda_e\, \Phi_{\text{energy}}(\mu), \quad \lambda, \lambda_e > 0,$$

and the constrained problem $\mu^\star \in \arg\min_{\mu\in\mathcal{S}} J(\mu)$. Then a minimizer $\mu^\star$ exists. Under strict convexity of $\mathrm{W}_2(\cdot, \mu^{\text{raw}})^2$ along appropriate Wasserstein geodesics on the feasible set, the minimizer is unique.

Proof sketch. Compactness of Ω implies tightness; $\mathcal{P}_2(\Omega)$ is W_2-complete. By W_2-continuity, $\mathcal{S}$ is closed. The objective is lower semicontinuous and coercive, so minimizing sequences are precompact and converge (direct method). Any limit point lies in $\mathcal{S}$ and attains the infimum [20, 23].

Remark 2 (Decoupling claim made explicit (A2))**.** Even if μ^{raw} is scheming or unsafe, the deployed law is the *projected* $\mu^\star \in \mathcal{S}$. When certification is statistically insufficient, the minimizer shifts toward safe baselines (refusal/no-op/handoff), yielding *graceful degradation* rather than unsafe competence.

4.4 The "How" of projection: finite classes and online control (A3, A6)

The theorem is infinite-dimensional; implementation is finite and auditable:

1. **Finite policy classes:** choose among a finite set of wrappers/policies $\{\pi_\theta\}_{\theta\in\Theta}$ (prompt wrappers, verifier strictness, tool permissions). Then projection reduces to discrete/parametric selection minimizing an empirical surrogate of J.
2. **Online adaptation:** implement projection as a feedback loop: monitors compute $\widehat{\Phi}_i$, and a controller adjusts θ (temperature, refusal thresholds, tool gates) to keep constraints satisfied. This aligns with online convex optimization and mirror descent-style updates [32, 33].
3. **Theory-to-practice transition justified:** the theory is posed on Ω for generality, but monitoring/certification occurs on finite prefixes Ω_T; the horizon T is a safety-critical engineering parameter defining the operational guarantee window.

5 Progressive Verification and Drift Detection (A3, A6)

Sequential testing provides operational safeguards.

Algorithm 1 Topology-Level Progressive Monitor (A3, A6)

```
Set thresholds θ_drift, budgets (E_max, τ), confidence δ
for t = 1, 2, ... do
    Receive progressive output y_t = (p_t, c_t, v_t, ℓ_t)
    if v_t = fail then
        Shutdown; record evidence bundle
    end if
    Estimate prefix law μ̂_t and compute drift statistic D_t = d(μ̂_t, μ̂_t^ref)
    if D_t > θ_drift then
        Shutdown; audit
    end if
    Update empirical risk bounds Φ̂_i^N and certify (Section 6)
    if certificate unknown then
        Degrade to refusal/no-op/handoff
    end if
end for
```

Remark 3 (Adversarial drift detection remains hard (A6))**.** Strategic adversaries may attempt to mimic baseline drift statistics. Defense-in-depth includes multiple independent monitors, randomized audit schedules, and cross-metric drift detection [26].

6 Computability: Conformal and PAC Certification (A6)

6.1 Bridging the computability gap via conformal bounds (A6)

Exact evaluation of $\Phi_i(\mu) = \mathbb{E}_\mu[r_i]$ over Ω is intractable in real time. We enforce a probabilistic safety certificate:

$$\mathbb{P}\Big(\Phi_i(\mu) \le \widehat{\Phi}_i^N + Q_{1-\delta}\Big) \ge 1-\delta,$$

where $\widehat{\Phi}_i^N$ is an empirical estimate from N trajectories/logs and $Q_{1-\delta}$ is a conformal calibration term [37, 38]. Operationally, accept only if $\widehat{\Phi}_i^N + Q_{1-\delta} \le \varepsilon_i$; otherwise degrade to safe no-op or human handoff (precautionary principle).

6.2 PAC sample complexity guarantee (A6)

Theorem 2 (PAC certification guarantee (A6))**.** Let Ω_T have bounded diameter under d_∞, and assume each Φ_i is L_i-Lipschitz w.r.t. W_2 on $\mathcal{P}_2(\Omega_T)$. Let $\hat{\mu}_N = \frac{1}{N}\sum_{j=1}^N \delta_{\omega^{(j)}}$ for i.i.d. samples $\omega^{(j)} \sim \mu$ truncated to horizon T. Then with probability at least $1-\delta$,

$$\sup_{i\in I} |\Phi_i(\hat{\mu}_N) - \Phi_i(\mu)| \le \left(\max_{i\in I} L_i\right) \mathsf{W}_2(\hat{\mu}_N, \mu) \le C_T \left(\max_{i\in I} L_i\right) \sqrt{\frac{\log(|I|/\delta)}{N}},$$

for a constant C_T depending on the diameter of Ω_T.

Remark 4 (Sample complexity)**.** To guarantee $\sup_i |\Phi_i(\hat{\mu}_N) - \Phi_i(\mu)| \le \varepsilon$ with confidence $1-\delta$, it suffices to take

$$N = O\left(\frac{C_T^2\, L_{\mathsf{max}}^2}{\varepsilon^2} \log \frac{|I|}{\delta}\right).$$

Sharper Wasserstein concentration bounds are available [39, 40].

7 Deceptive Alignment, Mesa-Optimization, Context Shift (A1, A6)

Empirical alignment-faking motivates not trusting internal reports [7]. Mesa-optimization risk is acknowledged [6].

Assumption 2 (Context stationarity (A6))**.** Let P_{cert} and P_{deploy} denote context distributions at certification and deployment. Assume $\mathsf{W}_2(P_{\mathsf{cert}}, P_{\mathsf{deploy}}) \le \delta_{\mathsf{context}}$. If violated, re-certification under deployment context is required.

Remark 5 (Trip-wire contexts (A6))**.** To reveal hidden objectives, include adversarial "trip-wire" contexts and require the progressive monitor to remain active in deployment.

8 Multi-Agent Extension and Its Limits (A5, A6)

Theorem 3 (Existence of aligned Nash equilibria (A5))**.** Assume each agent's feasible set is non-empty, compact, and convex in a locally convex topological vector space induced by weak convergence on probability measures, and utilities are continuous. Then an equilibrium exists [53, 54].

Remark 6 (Selection/enforcement remains open (A6))**.** Existence does not specify which equilibrium emerges; focal points and protocols matter [52]. In addition, collusion can occur without simple statistical dependence; causal threat models and intervention-based diagnostics can be required to distinguish coordination from confounding [18]. Cooperative-AI governance and

mechanism design identify open enforcement and coordination questions directly relevant to multi-agent alignment wrappers [17]. A concrete focal-point heuristic is to select (when multiple equilibria satisfy constraints) the equilibrium minimizing certified aggregate risk subject to feasibility, making the "least-risk" equilibrium salient.

9 Governance: Black Box Logging and Regulatory API (A6)

9.1 Tamper-evident logging (A6)

Use cryptographic commitments (hash chaining / Merkle trees) for auditability [45, 46].

9.2 Protocolized CertifiedCheck interface (A6)

Define a machine-parseable interface:

$$\texttt{CertifiedCheck}\big(\texttt{ruleset_hash}, \widehat{\mu}_t, \widehat{\Phi}^N, \varepsilon, \delta, \texttt{hashchain}\big) \to (\texttt{status}, \texttt{evidence_bundle}),$$

where:

- `ruleset_hash`: signed version identifier for AP and LTL rules,
- $\widehat{\mu}_t$: hash-committed estimate of prefix law,
- $\widehat{\Phi}^N$: calibrated conformal/PAC bounds,
- `hashchain`: commitments to logs and certificates.

Outputs:

- $\texttt{status} \in \{\texttt{CERTIFIED}, \texttt{NOT_CERTIFIED}, \texttt{UNKNOWN}\}$,
- `evidence_bundle`: machine-readable object with commitments to logs, certificates, and verifier outcomes.

10 Limitations and Open Problems (A6)

Despite theoretical advances, core practical challenges remain:

- **Specification problem not solved:** defining computable r_i, choosing ε_i, value pluralism, and critical vs soft laws remain central (made explicit and auditable, not eliminated). Process alignment can be anchored in AI risk management standards (e.g., change-controlled escalation/approval cycles) [16].
- **Computational feasibility:** high-dimensional drift metrics and real-time verification are unproven at millisecond scales; finite-horizon approximations introduce unquantified errors.
- **Engineering dependencies:** requires strict separation of generation from verification, privileged monitors, tamper-evident logging, and (optionally) resource enforcement hardware.
- **Adversarial robustness:** strategic manipulation of atoms/metrics/verifiers requires monitor hardening and defense-in-depth; adversarial ML motivates explicit threat models [47, 48]. Model-stealing and extraction risks motivate API-level controls and auditing [49, 50].
- **Log tampering and evidence integrity:** the practical security of `CertifiedCheck` depends on immutable log chaining and verifiable commitments; Merkle-tree / hash-chain constructions make ex post tampering detectable [45, 46].
- **Multi-agent scaling:** equilibrium existence does not address enforcement, incentive compatibility, or collusion-resistant monitoring.

11 Conclusion: The Axiomatic-to-Governance Pipeline (A1–A6)

This paper contributes an explicit end-to-end pipeline:

$$\begin{aligned}\textbf{Axioms (A1–A6)} &\to \textbf{Theory (topology, measure, game)}\\ &\to \textbf{Operationalization (LTL, progressive outputs, sequential monitors)}\\ &\to \textbf{Certification (conformal/PAC)}\\ &\to \textbf{Governance (black box, regulatory API)}.\end{aligned}$$

The key differentiator is A2: *alignment decoupled from training data quality* by projecting raw behavior laws onto an externally certified feasible set. In this framing, training may improve capability, but *deployment safety* is enforced by measurable constraints and external verification rather than by trusting internal narratives.

Synthesis of contributions. The mathematical core is a topology-on-behavior thesis: a deployed system induces $\mu \in \mathcal{P}(\Omega)$, and alignment is a closed feasibility constraint $\mu \in \mathcal{S}$ specified by scalar risk functionals Φ_i (A4) together with budgets and disclosure. The existence result (Section 4) establishes that, under explicit Wasserstein continuity/coercivity assumptions (A5), there exists an aligned projection $\mu^\star$ that is closest to the raw law μ^{raw} under a principled objective. The operational architecture (A3, A6) is *progressive verification*: the system emits filtration-aligned partial outputs and non-AI certificates so that violations appear as measurable drift in induced laws and can be handled by sequential tests with evidence bundles. The governance layer protocolizes these guarantees into a machine-parseable `CertifiedCheck` interface that can be audited and regulated.

Relation to existing alignment strategies. Where RLHF and Constitutional AI attempt to shape internal preferences and behavior through training incentives [9, 10], this work treats the trained model as potentially adversarial and makes alignment a property of *externally constrained deployment* (A1, A6). Debate-style approaches aim to elicit truth through interaction [11]; here the emphasis is instead on externally verifiable certificates, logging, and topological constraints on what action distributions may occur. CMDPs and shielding appear as special cases (constraints on behavior and runtime enforcement) [27, 28].

Limitations (honest closure). This framework does not eliminate the specification burden: it relocates it into explicit, versioned artifacts (risk scores, tolerances, LTL rulesets, and/or geometric anchors), which are *auditable* but still contested. Computational bottlenecks remain: high-dimensional drift metrics and real-time certification require approximations, and finite-horizon monitoring can introduce unquantified gaps. The tail-penalty bounds improve control of resource-evasion but do not, by themselves, prevent energy-orthogonal harms or strategic long-horizon planning. Multi-agent results establish existence but not enforceable equilibrium selection in open environments.

Concrete next steps. The most direct research agenda is empirical: (i) implement `CertifiedCheck` with tamper-evident logging and calibrated conformal/PAC certificates; (ii) benchmark drift monitors and certificate rates on realistic tool-using AI workflows; (iii) validate which metrics d and which Φ_i are stable under distribution shift; and (iv) develop red-team protocols for adversarial attempts to manipulate propositions, metrics, or certificates. In parallel, Appendix C outlines a complementary path that can reduce per-law specification by constraining the interaction space

itself via anchors and projections; this should be tested in controlled domains (e.g., financial advisory, compliance Q&A) before broader deployment.

Practical implications. Within the information-work scope, the proposal provides a regulator-compatible blueprint: external constraints, explicit certification confidence levels, and machine-parseable evidence bundles align naturally with risk management and governance regimes [13, 14]. The framework is thus positioned not as "alignment solved," but as a foundational, implementable scaffold that makes alignment claims precise, inspectable, and enforceable at deployment time.

A Appendix A: LTL Operationalization of the 20 Laws (A4, A6)

To operationalize laws as measurable functionals, we use LTL over atomic propositions derived from digital logs and runtime verification [35, 36].

Definition 3 (LTL safety specification)**.** For each Law i, define an LTL formula φ_i over atomic propositions AP:

- AP $= \{\text{harm}, \text{illegal}, \text{self_interest}, \text{in_context}, \ldots\}$,
- operators: $\Box$ (always), $\Diamond$ (eventually), $\circ$ (next), $\mathcal{U}$ (until),
- semantics: $\omega \models \varphi$ means trajectory ω satisfies φ.

Example 1 (Laws as LTL formulas)**.**

$$\begin{aligned} \text{Law 2 (no harm):} \quad & \varphi_2 = \Box\neg\text{harm}, \\ \text{Law 9 (context):} \quad & \varphi_9 = \Box(\text{task_active} \Rightarrow \text{in_context}), \\ \text{Law 11 (verification):} \quad & \varphi_{11} = \Box(\text{output} \Rightarrow \Diamond_{\leq k}\text{verified}). \end{aligned}$$

Definition 4 (Risk score construction)**.** Define

$$r_i(\omega) = \mathbb{1}_{\{\omega \not\models \varphi_i\}}.$$

For safety formulas $\Box\psi$, the violation set is a countable union of cylinder sets, hence measurable.

Proposition 2 (Measurability of LTL risk scores)**.** For any LTL formula φ over finite AP, the satisfaction set $\{\omega : \omega \models \varphi\}$ is $\mathcal{F}$-measurable and $r(\omega) = \mathbb{1}_{\{\omega \not\models \varphi\}}$ is measurable.

Proof sketch. Structural induction: atoms correspond to cylinder sets; Boolean operations preserve measurability; temporal operators yield countable unions/intersections of cylinder sets [35].

Remark 7 (Hybrid handling of subjective laws (A6))**.** Some laws (psychological harm, bias mitigation) resist clean Boolean atoms. Use hybrid enforcement: rule-based detectors plus statistical models with conservative escalation to human review. Under uncertainty, default to refusal or handoff. For bias specifically, operationalize a *process* (detection + review + corrective action) and maintain an auditable fairness criterion library (e.g., equality of opportunity templates) [51], embedded within a change-controlled risk governance cycle consistent with AI risk management practice [16].

Remark 8 (Governance for atom/ruleset updates (A6))**.** To prevent redefining atoms, AP and the LTL ruleset must be versioned, signed, and subject to change-control: each deployment binds to a ruleset hash; updates require regulator/auditor approval and re-certification.

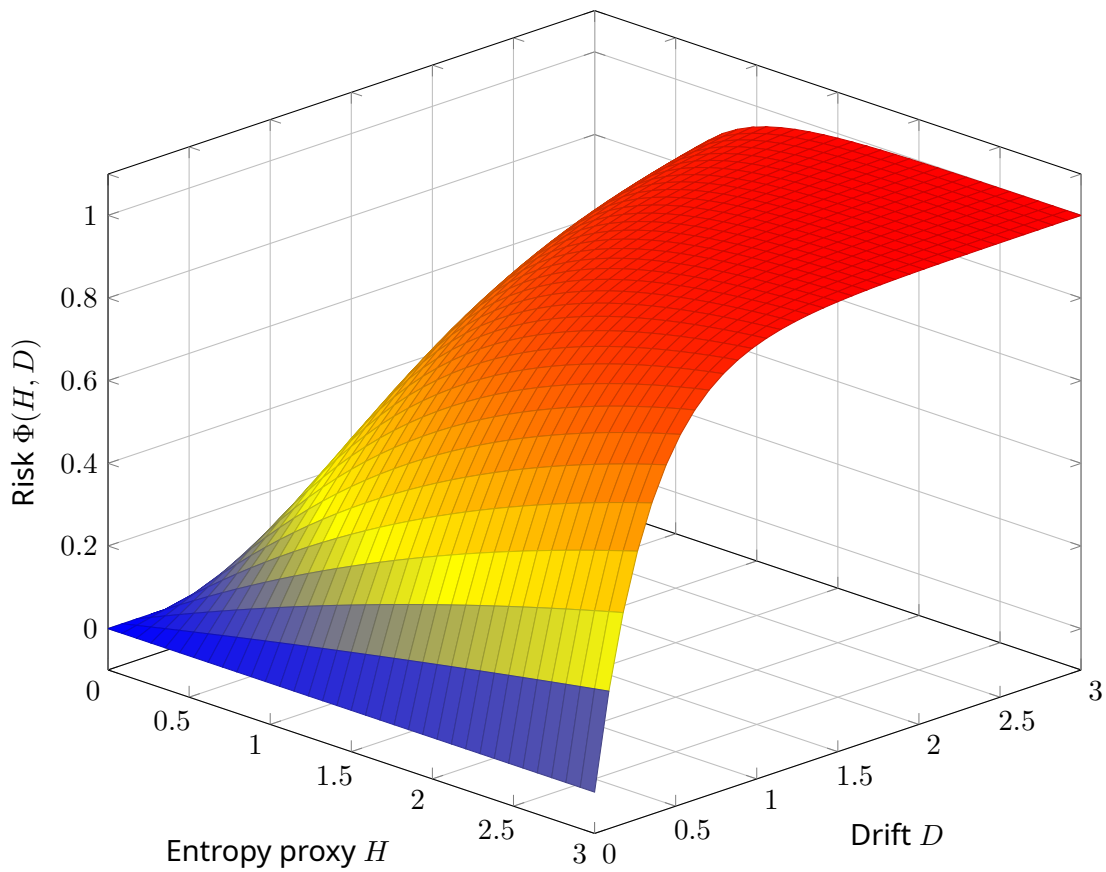

Figure 1: Illustrative risk surface. In deployment, Φ is computed from monitors/logs rather than chosen analytically.

B Appendix B: Illustrative Risk Surface (A4)

C Appendix C: Geometric / Topological Interaction Spaces with Anchors (A4, A6)

C.1 C.1 Interaction space as a finite product of infinite-dimensional components

To reduce per-law hand-coding of subjective r_i, we introduce a structured *interaction space* $\mathcal{X}$ that jointly represents (i) human inputs, (ii) AI outputs, and (iii) interaction artifacts, each living in an infinite-dimensional component space, but with only a *finite* number of components.

Let $m \in \mathbb{N}$ be small (case-specific), and let each component $\mathcal{Y}_k$ be a separable infinite-dimensional Hilbert space (e.g. $\ell^2(\mathbb{C})$) or a Banach space (e.g. ℓ^∞) with metric d_k. Define

$$\mathcal{X} := \prod_{k=1}^{m} \mathcal{Y}_k, \qquad d_{\mathcal{X}}(x, y) = \sum_{k=1}^{m} 2^{-k} d_k(x_k, y_k).$$

With m finite and each $\mathcal{Y}_k$ Polish, $\mathcal{X}$ is Polish, and $\mathcal{P}_2(\mathcal{X})$ supports Wasserstein control in the same spirit as the main $\mathcal{P}_2(\Omega)$ development (A5).

C.2 C.2 Constrained population of $\mathcal{X}$ via interfaces (human) and software (AI)

Human inputs are not allowed to "populate the space at will." Instead, users provide inputs through structured interfaces (forms, fields, bounded free-text boxes) with rate limits and schema

constraints. Let f_t denote form data at time t, and let E_H embed it into a low-codimension admissible region:

$$x_t^H = E_H(f_t) \in \mathcal{X}.$$

Similarly, the AI toolchain (xyz software) produces outputs embedded by E_A:

$$x_t^A = E_A(\text{output}_t) \in \mathcal{X}.$$

Unpredictable human behavior is represented operationally by increased submission rates, schema violations, or out-of-band residuals (below), which trigger escalation rather than unconstrained expansion.

C.3 C.3 Certified anchor library and dynamic "dummy" insertion

Let $\mathcal{D}_t = \{d_1, \ldots, d_{n_t}\} \subset \mathcal{X}$ be a *certified anchor (dummy) library* at time t, consisting of verified exemplars (e.g., known-correct instruction templates, disclosure templates, citation formats). The library is versioned and signed; changes are governed by the same change-control principles as rulesets (A6).

Dynamic insertion rule (informal but implementable): when a new object x is proposed (from human or AI), if its residual to the current admissible set exceeds tolerance (C.4), we can (i) reject/hand off, and/or (ii) propose a new candidate anchor from a pre-vetted candidate pool and require governance approval before activation. This "crowds" admissible regions over time while remaining auditable.

C.4 C.4 Projection-based admissibility and a topological residual functional

Let $\mathcal{K}_t \subset \mathcal{X}$ denote the admissible set induced by anchors and constraints (e.g., cones, norm balls, schema-valid manifolds). In the Hilbert setting with closed convex $\mathcal{K}_t$, the metric projection $P_{\mathcal{K}_t}(x)$ is well-defined and unique; in more general Banach settings, one uses proximal maps or constrained optimization [23, 34].

Define the residual:

$$\text{res}_t(x) := \|x - P_{\mathcal{K}_t}(x)\|,$$

where $\|\cdot\|$ is the task-calibrated norm on $\mathcal{X}$ (chosen to reflect operational semantics, not generic embedding similarity). An admissibility rule is:

$$\text{res}_t(x) \le \varepsilon_{\text{topo}} \quad \Rightarrow \quad \texttt{ACCEPT}; \qquad \text{res}_t(x) > \varepsilon_{\text{topo}} \quad \Rightarrow \quad \texttt{HANDOFF/REFUSE}.$$

This induces a new risk functional compatible with the main framework:

$$\Phi_{\text{topo}}(\mu) := \mathbb{E}_{x\sim\mu}\big[\min\{1, \text{res}_t(x)/\varepsilon_{\text{topo}}\}\big] \le \tau_{\text{topo}}.$$

The intent is *not* to claim that geometry "solves" ethics, but to provide a controllable, auditable safety envelope that can reduce manual rule-writing for regimes where uncertainty is high.

C.5 C.5 Human unpredictability as controlled load and residual drift

A naive density count can be misleading. A more robust operational signal is the joint behavior of (i) submission rate, (ii) schema violation rate, and (iii) residual drift statistics. For example, track

$$\widehat{p}_t := \frac{\#\{\text{schema violations up to } t\}}{\#\{\text{submissions up to } t\}}, \qquad \widehat{r}_t := \text{running mean of res}_t(x).$$

Escalate when either exceeds thresholds, or when residual distributions drift relative to a certified baseline (A3/A6), consistent with the drift-monitor logic in the main text.

C.6 C.6 Concrete worked example: financial advisory chatbot (A2, A4, A6)

Worked example: financial advisory chatbot. Consider $m = 3$ components: $\mathcal{Y}_1$ (user queries), $\mathcal{Y}_2$ (advice outputs), $\mathcal{Y}_3$ (citation/source space).

Anchor library. A certified anchor library $\mathcal{D}$ includes (illustratively):

- d_1: "Conservative portfolio allocation" template,
- d_2: "Risk disclosure" template (jurisdictional),
- d_3: "Market data citation format" template,
- ...(10–20 certified exemplars, versioned and signed).

Constraints. Define an admissible set $\mathcal{K}$ via constraints such as:

- Norm bounds: $\|x_2\|_2 \leq M_{\text{advice}}$ (prevent extreme recommendations),
- Source sparsity / diversity proxy: $\|x_3\|_0 \leq k_{\text{sources}}$ together with minimum-source requirements (enforced via schema, not only norms),
- Permission cone: x_3 must draw only from approved data APIs (enforced by tool-gating + logging).

User query and projection gate. User submits via a form: "Should I invest my retirement in crypto?" The embedding E_H maps to $x \in \mathcal{X}$. The system computes $P_{\mathcal{K}}(x)$:

- If $\|x - P_{\mathcal{K}}(x)\| \leq \varepsilon_{\text{topo}}$: accept; generate a constrained response with required disclosures and sources.
- Else: "Your query requires specialized advice. Connecting to a human advisor..." (graceful handoff).

Example outcome and logging. Suppose $\|x - P_{\mathcal{K}}(x)\| = 2.3 > \varepsilon_{\text{topo}}$. The system hands off. Log records include: query hash, residual, handoff decision, confidence parameters, and a Merkle/hash-chain commitment ℓ_t so post-hoc tampering is detectable [45, 46]. The evidence bundle ties back to the `CertifiedCheck` contract.

C.7 C.7 Trade-offs and failure modes (honest analysis)

What are "good anchors"? Anchors are themselves a specification artifact: the difficulty shifts from writing per-law r_i to curating $\mathcal{D}$ and $\mathcal{K}$ (templates, schemas, cones). This is a *relocation* of burden, not elimination. The advantage is that anchors can be versioned, audited, tested, and governed.

Convexity is fragile. Safety semantics are often non-convex; closed-convex $\mathcal{K}$ is a controllable *envelope*, not a full semantic truth set. In practice one combines convex envelopes with discrete rule checks and escalation paths (Appendix A), rather than claiming convexity captures safety fully.

Projection semantics depend on representation. Distances in generic text embeddings need not correlate with safety. The norm in res_t must be defined on *structured representations* (schemas, claim-citation graphs, tool-call traces, disclosure objects) whose deviations are operationally meaningful and testable, not on uncontrolled latent vectors.

Governance latency. If the task regime drifts, anchor updates require a governance epoch. Mitigation is conservative defaults (handoff/refusal), small incremental updates, and continuous monitoring/certification cycles rather than infrequent large changes.

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